Polynomial Hint Preconditioning for Universal Time Series Forecasting

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−2.9%
geometric-mean MASE improvement over baseline

329/485
paired wins ($p < 10^{-14}$, Holm-corrected)

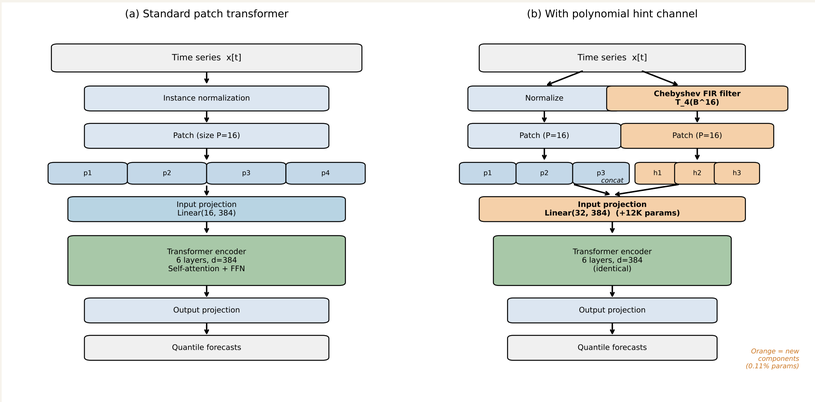
12K
extra parameters (0.11% overhead)

11.4M
Moirai-2 Small parameters, 5 seeds, 97 configs

1 — Problem: The Cross-Patch Bottleneck

Universal time series forecasters *patch* the input into blocks of P samples and embed each patch independently before any attention layer. A daily cycle in 15-minute data spans 96 steps = 6 patches—**attention alone must discover this**.

Smaller models, with fewer heads/layers, are most constrained. We inject cross-patch information *before the first attention layer* via a fixed polynomial filter that preserves parameter count.



2 — Our Method: Hint Preconditioning

Compute the **hint channel**

$$h[t] \;=\; \sum_{k=0}^{d} a_k\, x[t-ks], \qquad s = P,$$

and *concatenate* it with the target and observation mask before the per-patch projection:

$$\mathbf{e}_i = \mathrm{ResBlock}_{3P\to d}([\mathbf{p}_i;\, \mathbf{m}_i;\, \tilde{\mathbf{h}}_i]).$$

For $d = 4$, $s = 16$:

$$h[t] \;=\; x[t] - x[t-32] + \tfrac{1}{8}\, x[t-64].$$

Only the first linear layer widens; the transformer encoder is unchanged.

Hint dropout. At training each patch's hint is zeroed with prob. 0.1; at inference the full hint is provided.

3 — Why Chebyshev? (Theory)

Two classical results motivate the choice of filter coefficients.

(i) Chebyshev minimax theorem. Among all monic polynomials of degree d,

$$\tilde{T}_d \;=\; \arg\min_{q \text{ monic}} \max_{x\in[-1,1]} |q(x)|, \quad \|\tilde{T}_d\|_\infty = 2^{1-d}.$$

(ii) LDS preconditioning (Marsden & Hazan '25): applying $p(B)$ to an observation sequence $y_t = CA^t h_0$ is equivalent to multiplying the hidden dynamics by $p^*(A)$. The Chebyshev choice minimizes the worst-case spectral amplification.

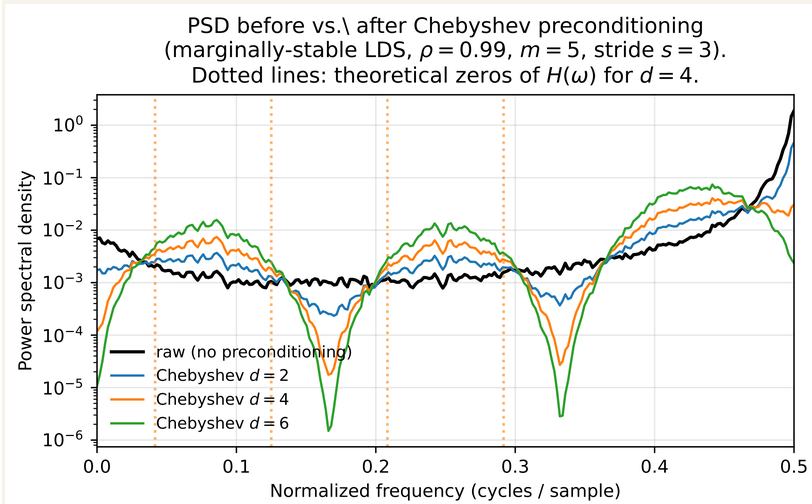
Frequency response (new). For our filter at stride s the magnitude

$$|H(\omega)|^2 = p\!\left(e^{-j2\pi s\omega}\right) \overline{p\!\left(e^{-j2\pi s\omega}\right)}$$

is a real trigonometric polynomial in $\cos(2\pi s\omega)$, and attains its minimum value

$$|H(\omega_k)| \;=\; 2^{1-d} \quad \text{at} \quad \omega_k = k/(2s),\; k \in \mathbb{Z}.$$

4 — Synthetic LDS: Theory Validation



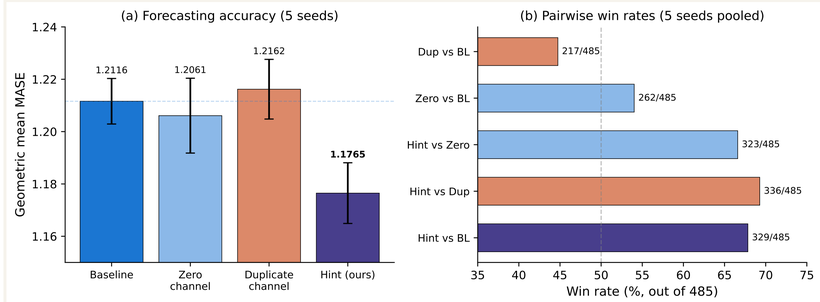
Simulated LDS ($\rho = 0.99$, $m = 5$, stride $s = 3$).
Filter notches land at $\omega = 0, 1/6, 1/3$—**exactly where the theorem predicts**. Notch depth scales as 2^{2-2d}, matching the Chebyshev minimax constant.

5 — Empirical Results

GIFT-Eval: 97 dataset × horizon configurations, context length 4,000, 5 seeds.

Method	MASE	Δ%	Wins / 485
HD10 (ours)	**0.837**	**−2.9%**	**329**
Zero (capacity ctrl)	0.858	−0.5%	262
Baseline	0.862	—	—
Duplicate (cap. ctrl)	0.865	+0.4%	217

The capacity controls hurt. Duplicating the input as a second channel wins only 44.7% of the time (worse than chance); zeros are neutral. *Polynomial content drives the improvement, not extra parameters.*



6 — Statistical Methodology (CSML)

- Paired sign test, 485 observations: $p = 2.9 \times 10^{-15}$.
- **Holm–Bonferroni** per-seed: 4 of 5 seeds individually significant at $\alpha = 0.05$.
- **BCa 95% bootstrap** ($B = 5{,}000$): relative improvement $\in [-3.71\%, -2.12\%]$; excludes zero.
- Cohen's $d = +0.31$ (paired log-MASE); Cliff's $\delta = +0.36$.
- Post-hoc power: $n = 65$ suffices for 80% power; we have 485 (7.5×).

Takeaways

1. Theory meets practice. Chebyshev preconditioning is *minimax-optimal* for spectral compression; empirically, it beats all tested alternatives (EMA, Legendre, finite differences, duplicate, zero).

2. Capacity-efficient. 0.11% extra parameters yield −2.9% MASE and −3.9% at 100K steps. No change to the transformer encoder.

3. Predictable from spectrum. Datasets with more spectral mass in the filter's passband benefit more ($r = +0.10$, 39 datasets)—mechanism consistent with Theorem 2.